

The PAC model can be characterised by

$$(\mathcal{X}, \mathcal{Y}, \mathcal{C}, \mathcal{D}, \mathcal{H}, S)$$

where $\mathcal{X}$ is the input space, $\mathcal{Y}$ is the output space, $\mathcal{C}$ is the concept class, $\mathcal{D}$ is the distribution over the input space, $\mathcal{H}$ is the hypothesis class, and m is the sample size.

The PAC model is a framework for understanding the learnability of a concept class $\mathcal{C}$ given a labelled sample $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$ drawn i.i.d. from a distribution $\mathcal{D}$ over $\mathcal{X}$. The goal is to find a hypothesis $h_S : \mathcal{X} \rightarrow \mathcal{Y} \in \mathcal{H}$ that approximates the target concept $c : \mathcal{X} \rightarrow \mathcal{Y} \in \mathcal{C}$ with small *generalization error* denoted by $L_{\mathcal{D},c}(h)$.

Definition: Generalization error

Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$, the generalization error is defined as

$$L_{\mathcal{D},c}(h) = \mathbb{P}_{x \sim \mathcal{D}}[h(x) \neq c(x)] = \mathbb{E}_{x \sim \mathcal{D}}[\mathbb{I}(h(x) \neq c(x))]$$

The generalization error is not directly accessible to the learner, as both $\mathcal{D}$ and c are unknown.

Definition: Empirical error

Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a labelled sample $S = \{(x_i, y_i)\}_{i=1}^m$ drawn from the distribution $\mathcal{D}$, the empirical error is defined as

$$L_{S,c}(h) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(h(x_i) \neq c(x_i)) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(h(x_i) \neq y_i)$$

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